Networks

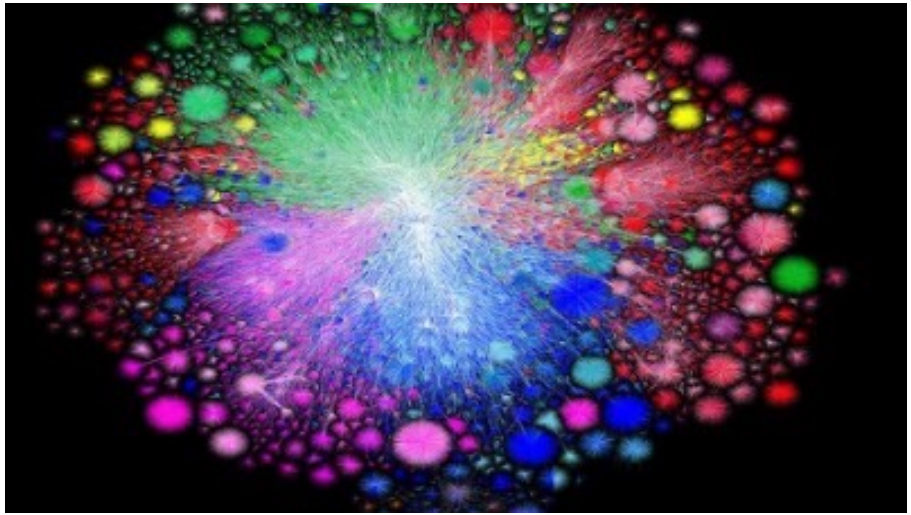
COMP7507
Visualization & Visual Analytics



SCHOOL OF
COMPUTING &
DATA SCIENCE
The University of Hong Kong

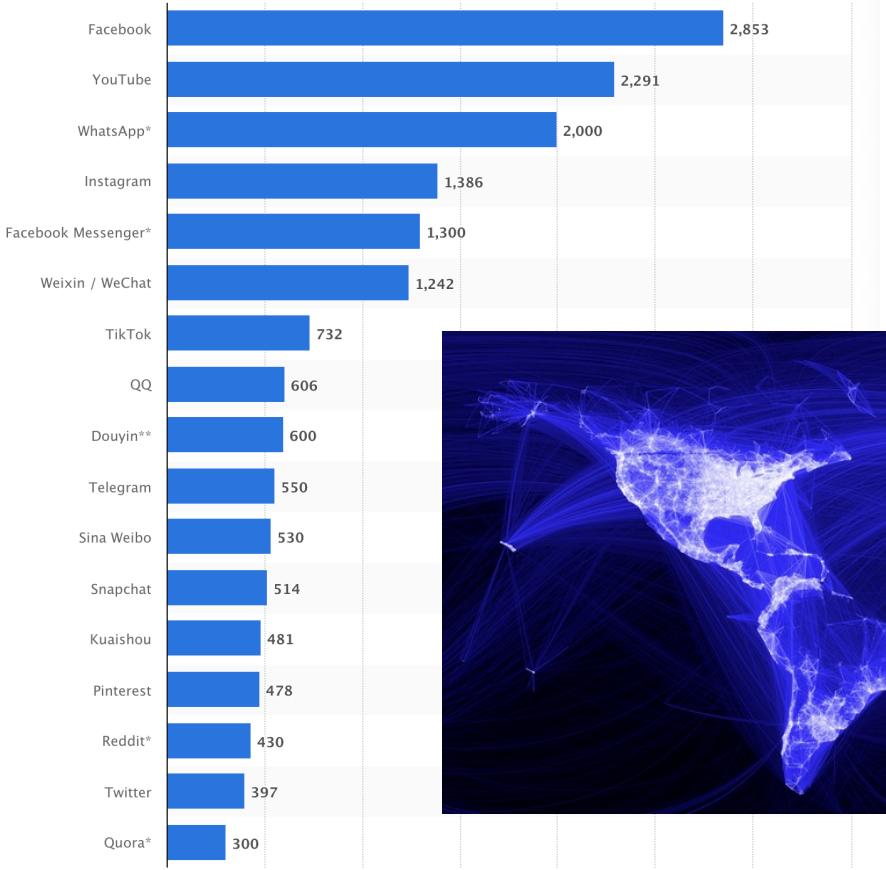
Network Size

- Size of the WWW: ~40 billion pages (http://www.worldwidewebsize.com/)



[Opte project (opte.org)]

Network size



[https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/]

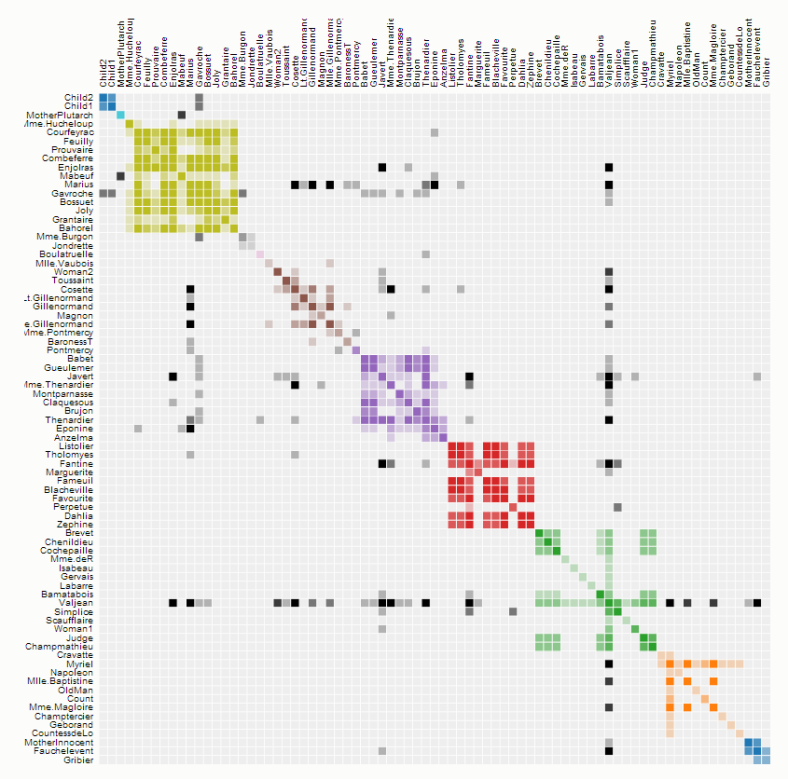
[Paul Butler, http://paulbutler.org/archives/visualizing-facebook-friends/]



Graph Drawing

- Direct calculation based on graph structure
  - Adjacency matrix layout
- Optimization-based
  - Optimizing the graph aesthetic constraints
  - Force-directed layout

Adjacency Matrices

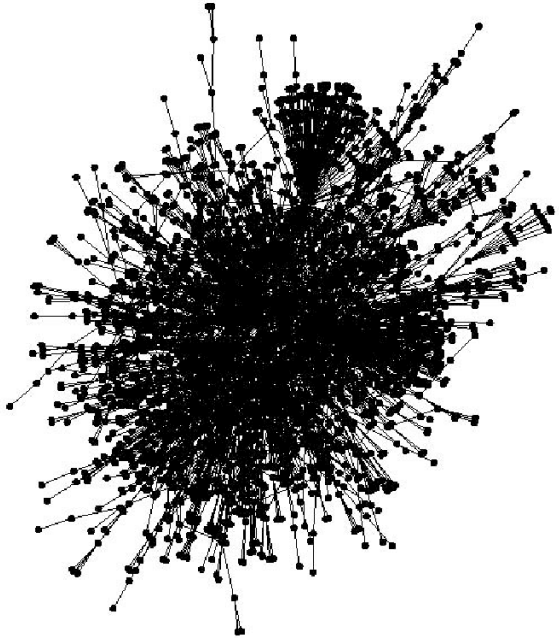


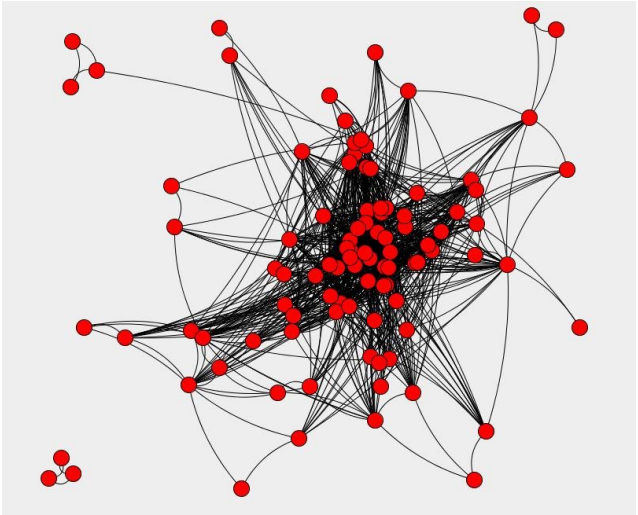
Les Misérables Co-occurrence

[http://bost.ocks.org/mike/miserables/]

Node-Link Layout

- Severe edge crossings and cluttering

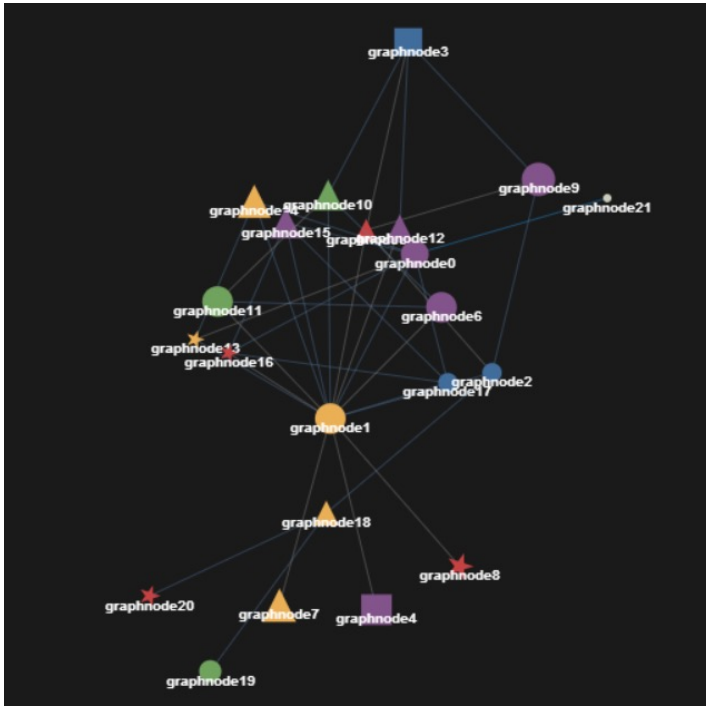




Optimization Techniques

- Formulate the layout problem as an optimization problem
- Governed by an equation incorporating
  - the costs of items (e.g., distance between nodes) to be optimized; and
  - some constraints (e.g., no edge crossings allowed)
- Non-deterministic and therefore resulting layout is unpredictable
- **Force-Directed Layout** is commonly used for undirected graph

Force Directed Layout



[http://philogb.github.io/jit/static/v20/Jit/Examples/ForceDirected/example1.html]

Force Directed Layout

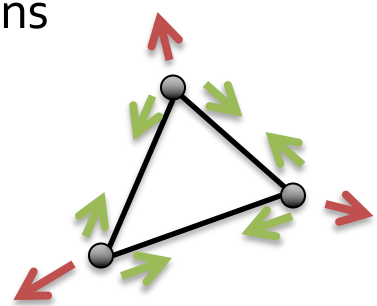
- To model nodes and edges of a graph as physical bodies tied with springs
- Two principles:
  - Vertices connected by an edge should be drawn near each other.
  - Vertices should not be drawn too close to each other.
- Edges = springs (attractive force)
  - Hooke's Law: $F = k * x$

  k: constant; x: length of spring
- Nodes = charged particles (repulsive force)
  - Coulomb's Law: $F = -k * q_1 * q_2 / x^2$

  k: constant; x: distance between two particles; q_1,q_2: charges

Force Directed Layout

- Initialize node positions (e.g., random placement)
- Do this iteratively until stabilized:
  1. Calculate forces for nodes
  2. Update node positions

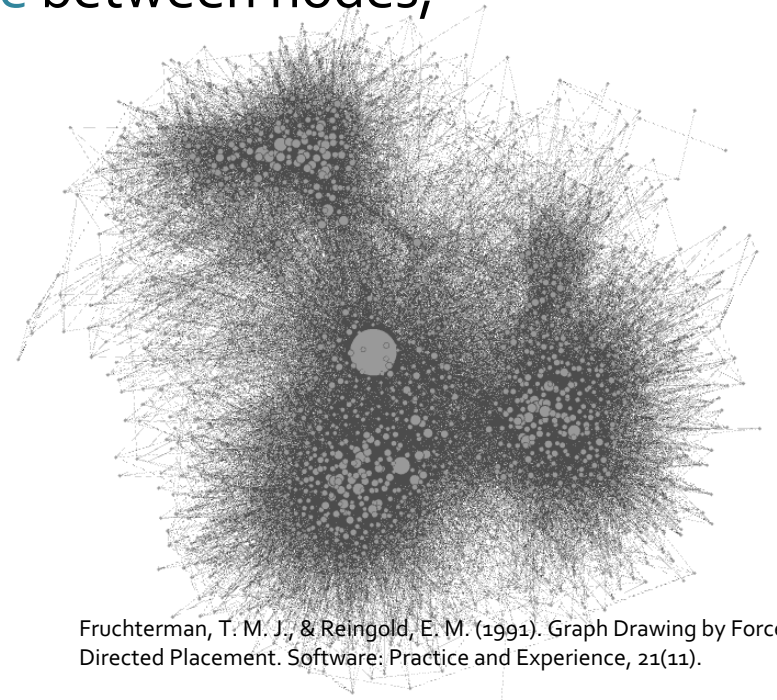


- Different physical models lead to layouts of varying quality

Fruchterman-Reingold Algorithm

- A force directed layout algorithm
- Calculate an optimal distance between nodes,
  $\delta = C * \text{sqrt}(\text{area} / |V|)$,
  where C is a constant
- Attractive force
  $f_a(x) = d^2 / \delta$
- Repulsive force
  $f_r(x) = -\delta^2 / d$

d: distance between two nodes



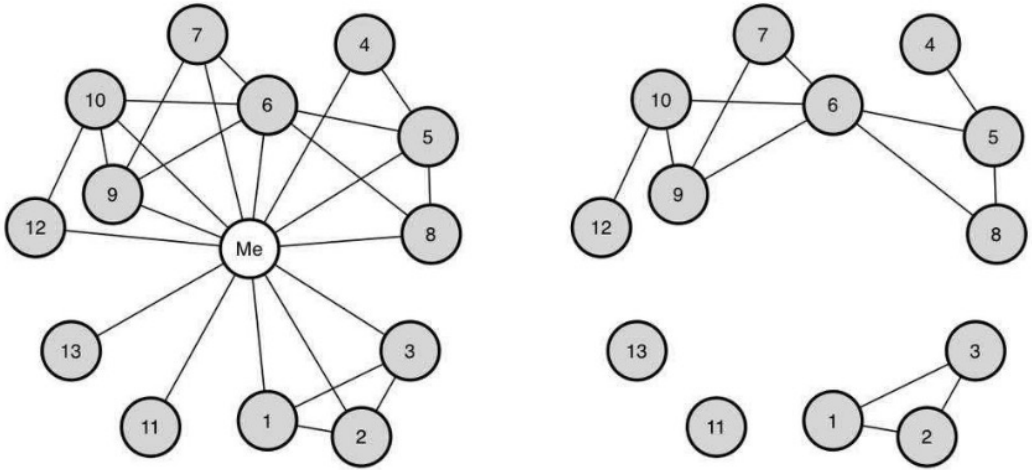
Fruchterman, T. M. J., & Reingold, E. M. (1991). Graph Drawing by Force-Directed Placement. Software: Practice and Experience, 21(11).

Filtering

- To reduce the number of visible elements in a graph being viewed
  - Improves clarity
  - Increases performance of layout and rendering
- Removing some “redundant” edges to better reveal network topology, and to facilitate the clustering process

Filtering

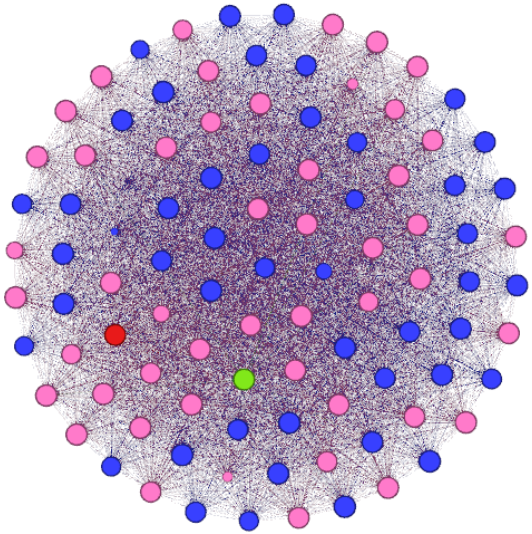
- E.g., Edges linking you and your friends in your Facebook network
- Ego network



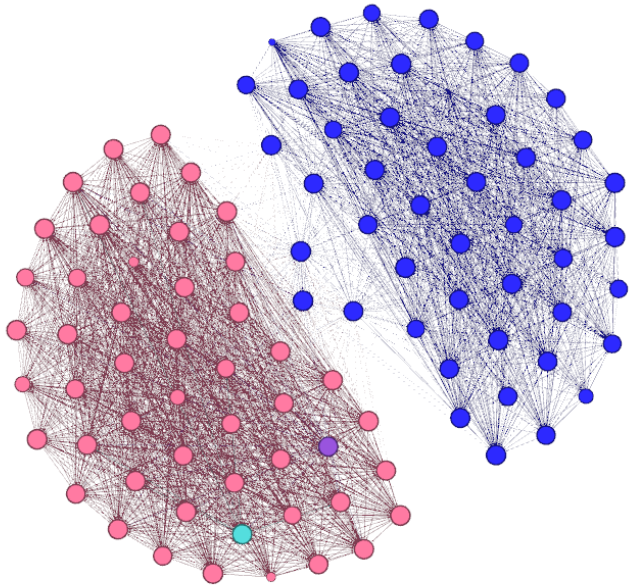
[Hansen 2011]

Filtering

Results of Fruchterman-Reingold layout on
the 2007 US Senate voting data with and without edge filtering



Without filtering



Edges corresponding to
% agreement < 0.65 are filtered

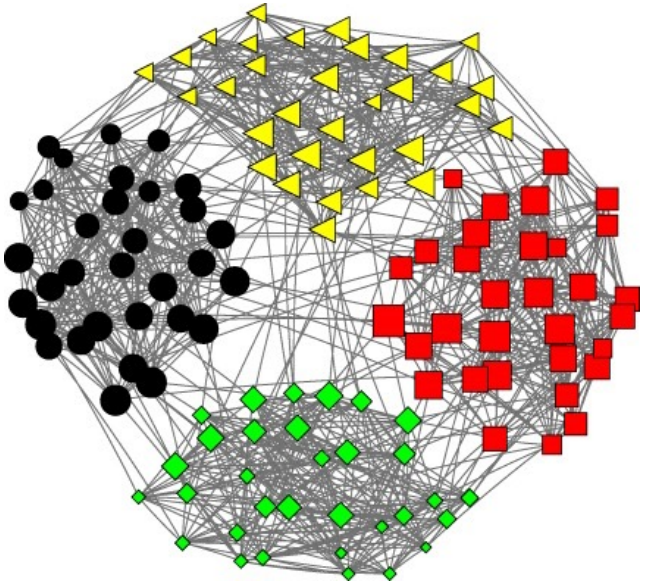
Node color represents party affiliation of a senator

Clustering

- Structure-based clustering
  - Use only structural information (e.g., edge connections) of a graph
- Content-based clustering
  - Use semantic data associated with graph elements
  - Application specific

Clustering

- It is natural to choose the clustering with the least number of edges between members
  - or with the minimum total weight of the edges connecting members for graphs with weighted edges
- Force directed layout algorithms can also form clusters naturally

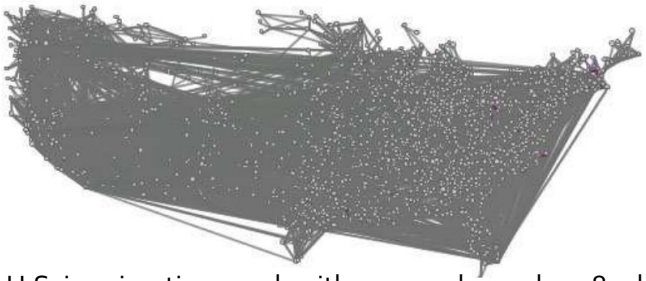


Edge Bundling

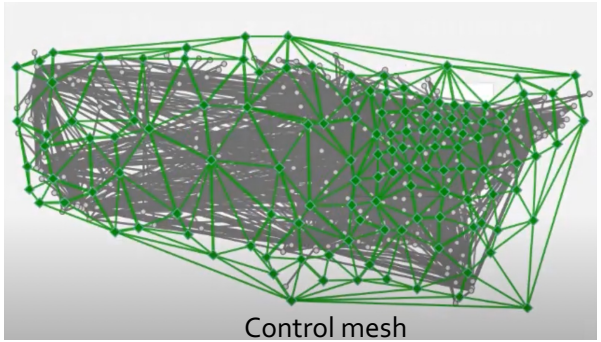
- Clustering of edges instead of nodes
- To reduce cluttering
- Examples:
  - Geometry-Based Edge Clustering
  - Hierarchical Edge Bundles

Geometry-Based Edge Clustering

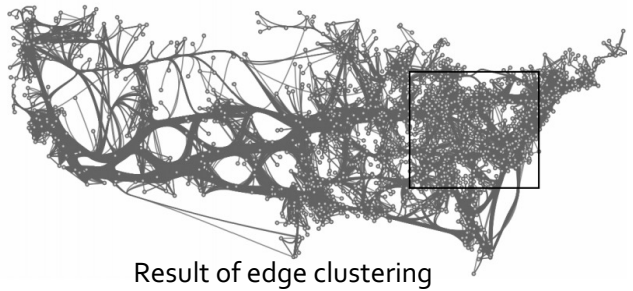
- A control mesh is generated based on the underlying graph structure (node position + edge distribution), to guide the edge clustering process



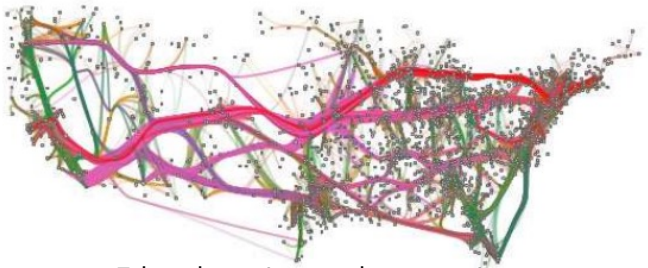
U.S. immigration graph with 1790 nodes and 9798 edges.



Control mesh



Result of edge clustering



Edge clustering + color + opacity

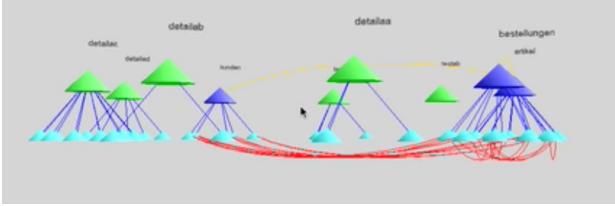
[https://youtu.be/hourhUK6Sb4?t=50]

[Cui et al., "Geometry-Based Edge Clustering for Graph Visualization," *TVCG*, 2008]

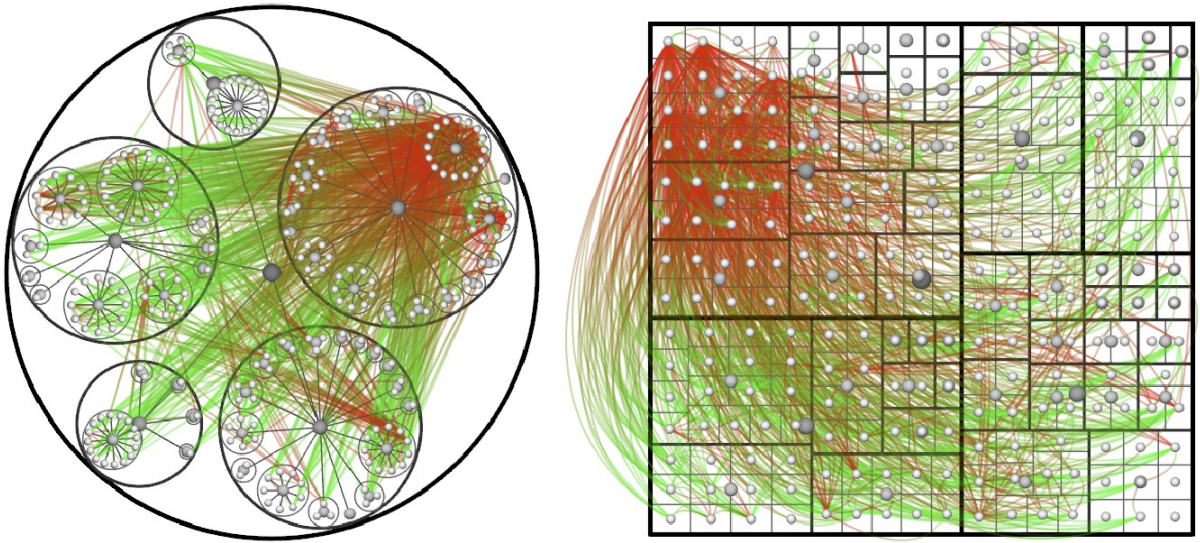
Hierarchical Edge Bundles

- Some data sets are with both non-hierarchical (adjacency) and hierarchical relations.
  - Source codes for a software:
    - Adjacency: dependencies of classes
    - Hierarchical: directories -> files -> classes
  - Social networks:
    - Adjacency : if people are acquainted and nature of relations
    - Hierarchical: circles or groups of people -> individuals
  - Citation networks:
    - Adjacency: citations among publications
    - Hierarchical: institutions -> departments -> publications

Hierarchical Edge Bundles



- Visualization examples without edge bundling



Colored edges representing adjacency relations on (left) balloon trees, and (right) tree maps.

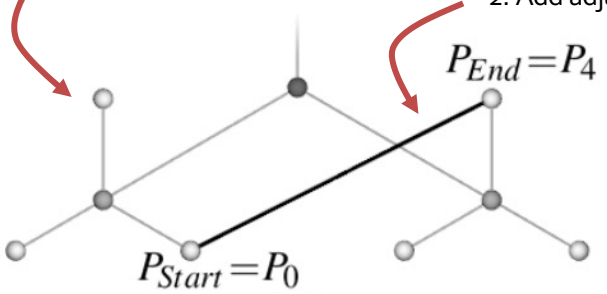
[Holten, "Hierarchical Edge Bundles: Visualization of Adjacency Relations in Hierarchical Data," *TVCG* , 2006]

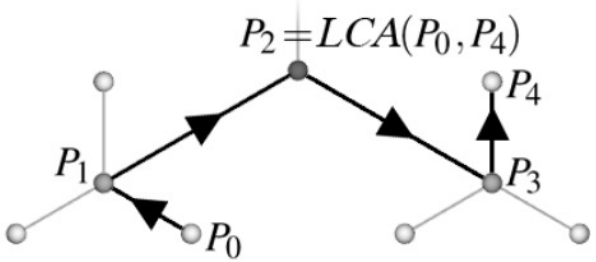
Hierarchical Edge Bundles

- Bundle adjacency edges along tree hierarchy

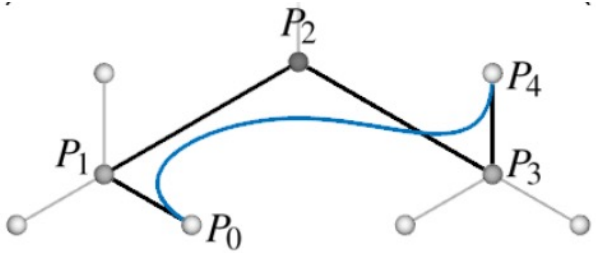
1. Tree drawing for hierarchical relations

2. Add adjacency edge (Po, P4) to graph





Find path that goes through least common ancestor (LCA)

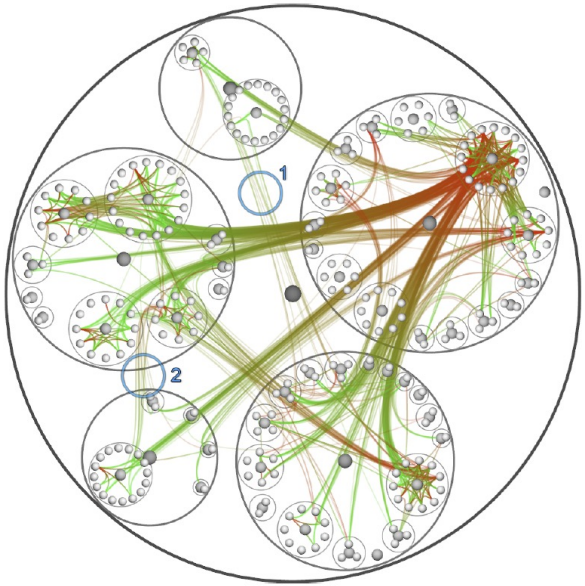


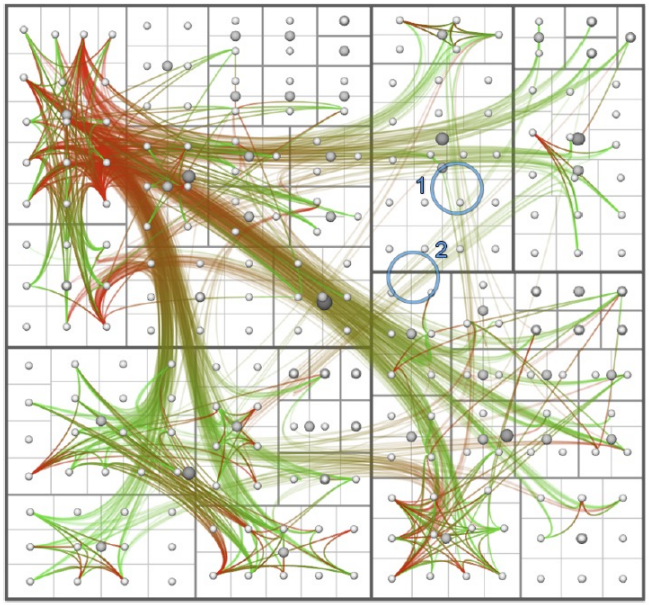
Draw curved adjacency edge that "aligns" to tree

(cf. Bezier curve)

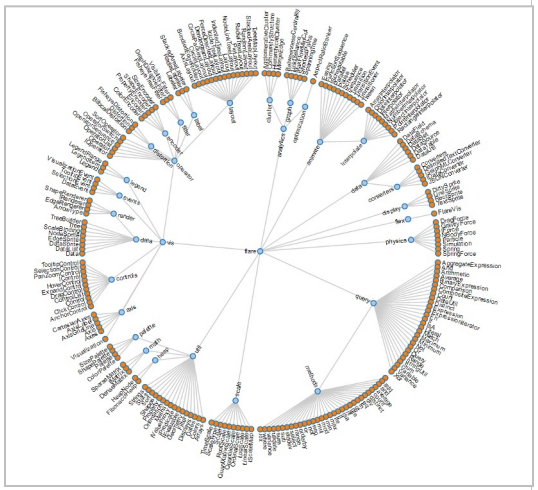
Hierarchical Edge Bundles

- Examples with hierarchical edge bundling

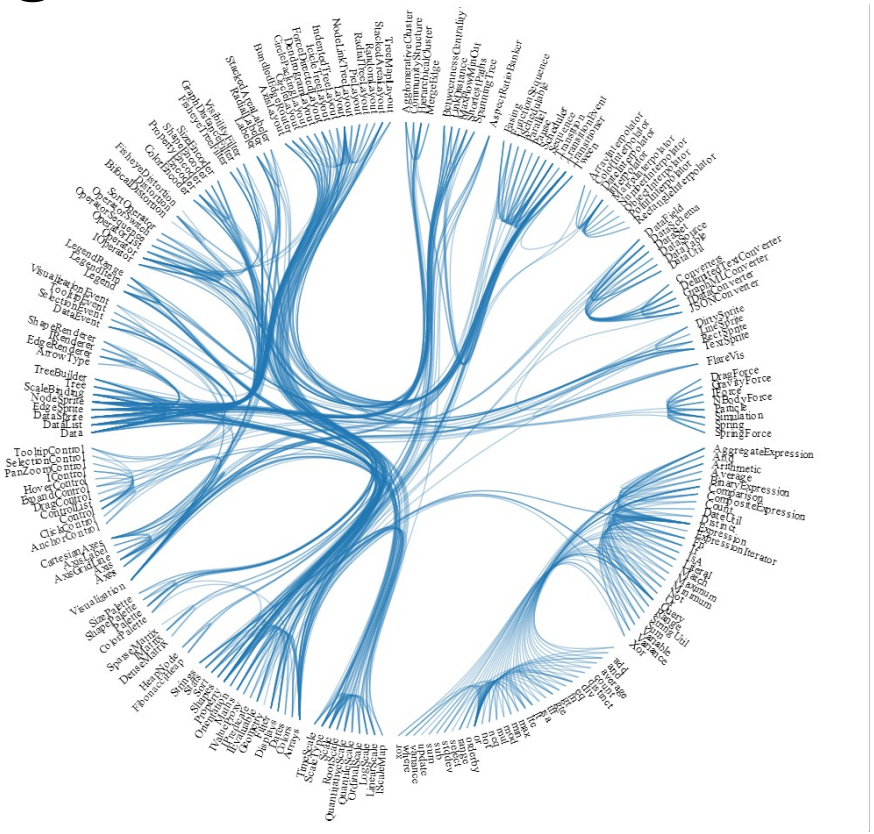




Hierarchical Edge Bundles



Tree structure is used only for edge clustering and is not shown (cf. "Trees", p.21)



[http://mbostock.github.io/d3/talk/20111116/bundle.html]

Understanding a Network

- Visualization alone cannot provide full understanding of a graph or network
- **Network graph metrics** are quantitative measures for describing a network, characterizing a subgroup or some specific nodes within a network
  - Influential people in a social network (e.g., celebrities in Twitter or Weibo)
  - Gatekeepers connecting communities (e.g., for headhunting in LinkedIn)

Understanding a Network

- Studying network metrics over time helps understand how a network evolves
- Network metrics help improve the visual display of a network
  - By assigning different visual attributes to the nodes
  - By filtering and showing only the important nodes

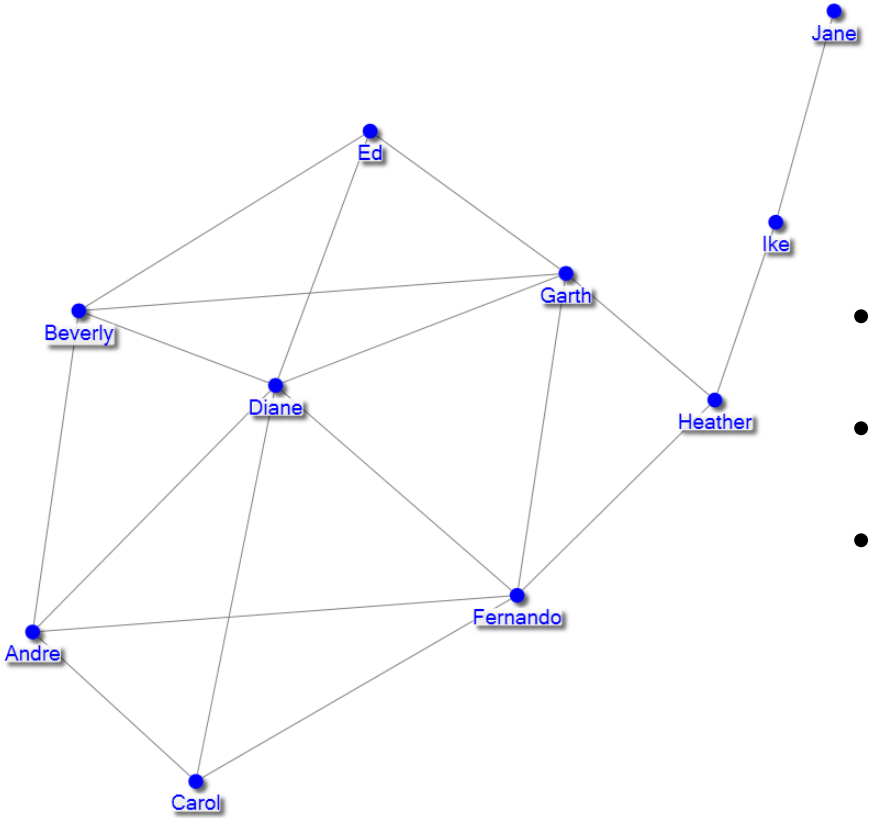
Network Metrics

- **Overall graph metrics**
  - Graph type; # of vertices; # of edges
  - Self loops (e.g., a person replying his own emails)
  - Connected components
  - Isolated vertices
  - Maximum geodesic distance (aka diameter)
    - i.e., the distance between two nodes that are farthest apart
  - Average geodesic distance
    - i.e., average distance from one node to another through the graph edges
  - Graph density
    - i.e., # edges ÷ # of possible edges
  - etc.

For undirected graph, graph density = (2 * |E|) / (|V| * (|V|-1))

For directed graph, graph density = |E| / (|V| * (|V|-1))

Overall Graph Metrics



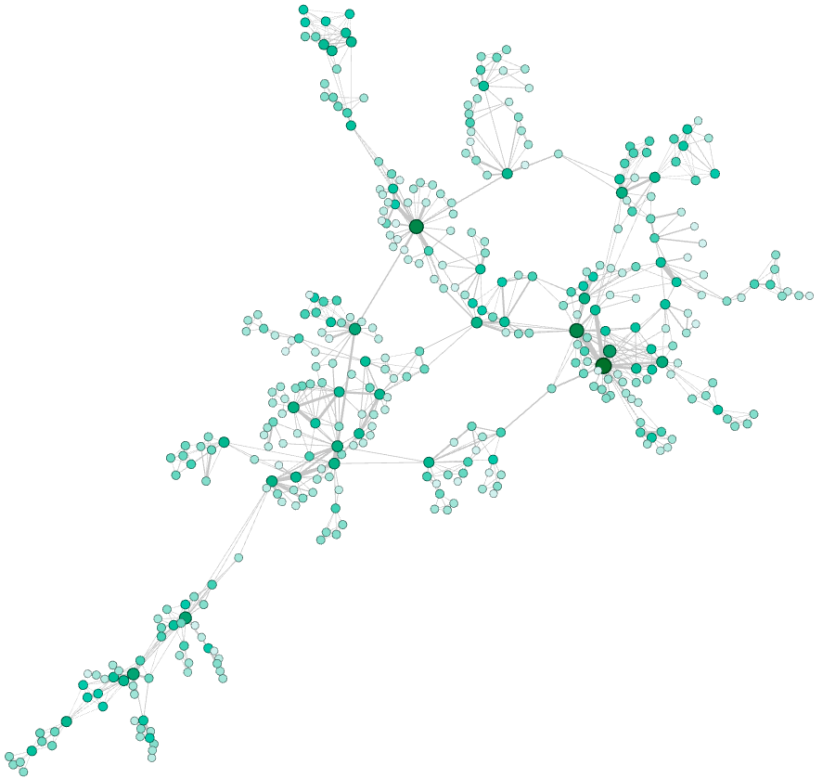
- Max. geodesic dist. = 4
- Avg. geodesic dist. = 1.98
- Graph density = 18/45 = 0.4

Network Metrics

- **Node metrics**: structure-based metrics associated with a node
  - Degree (in-degree, out-degree)
  - Betweenness centrality
  - Closeness centrality
  - Clustering coefficient
  - Eigenvector centrality
  - PageRank
- Use for **identifying special or important nodes** or subgroups
- There are **edge metrics** as well (e.g., edge betweenness)

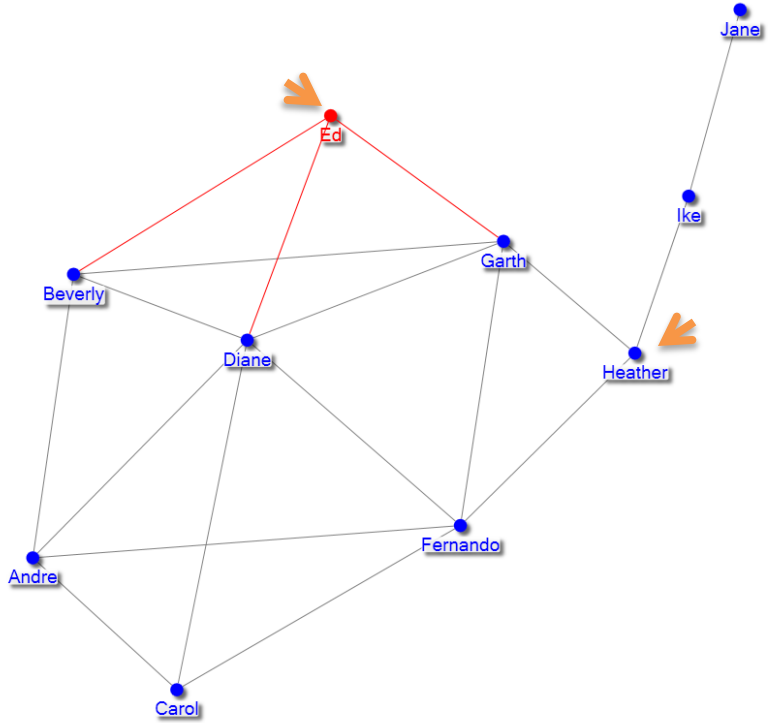
Degree Centrality

- A node is considered more important if it is of a greater degree
- **Centrality** means "Importance"



Betweenness Centrality

- Measures the importance of a person in passing information within a network



Who is more important in terms of betweenness, Ed or Heather?

Intuitively, Heather is of higher betweenness centrality than Ed because removing Heather results in isolated nodes

Betweenness Centrality

- The **betweenness centrality** of a node:

$$C(v) = \sum_{s,t \neq v \in V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

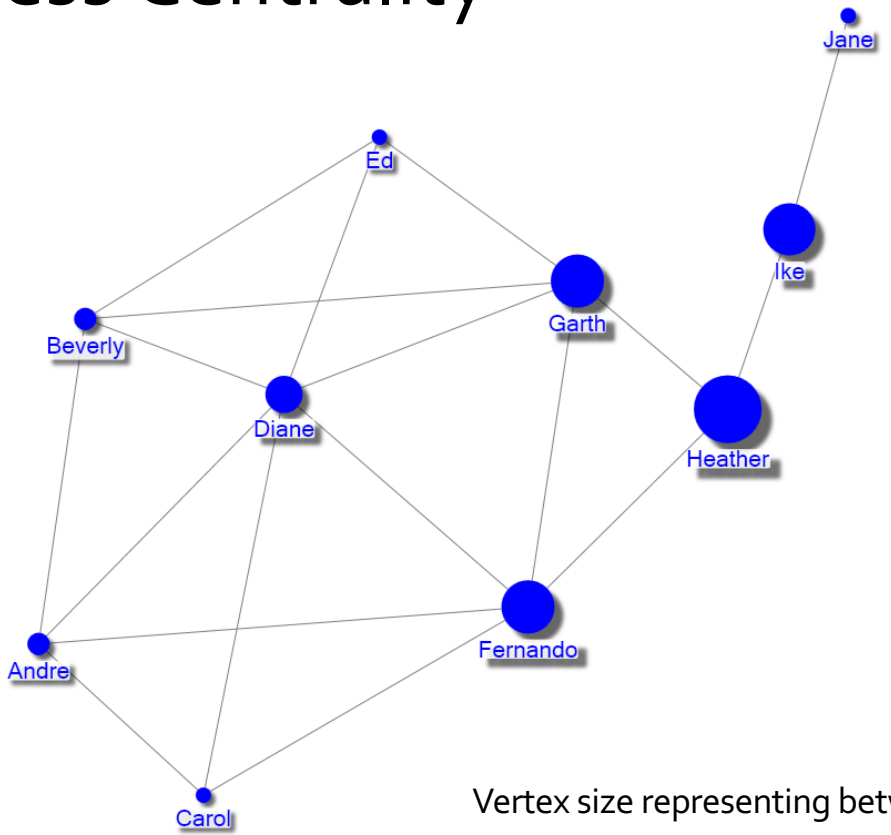
$\sigma_{st}(v)$: # of shortest path between *s* and *t* passing through *v*

σ_{st}: Total # of shortest path between *s* and *t*

- Equivalent to computing the all-pairs shortest path of a graph — Complexity: $O(|V|^3)$

O(|V||E|) on unweighted sparse graph
Ulrik Brandes, "A Faster Algorithm for Betweenness Centrality ", Journal of Mathematical Sociology, 2001.

Betweenness Centrality



Vertex size representing betweenness

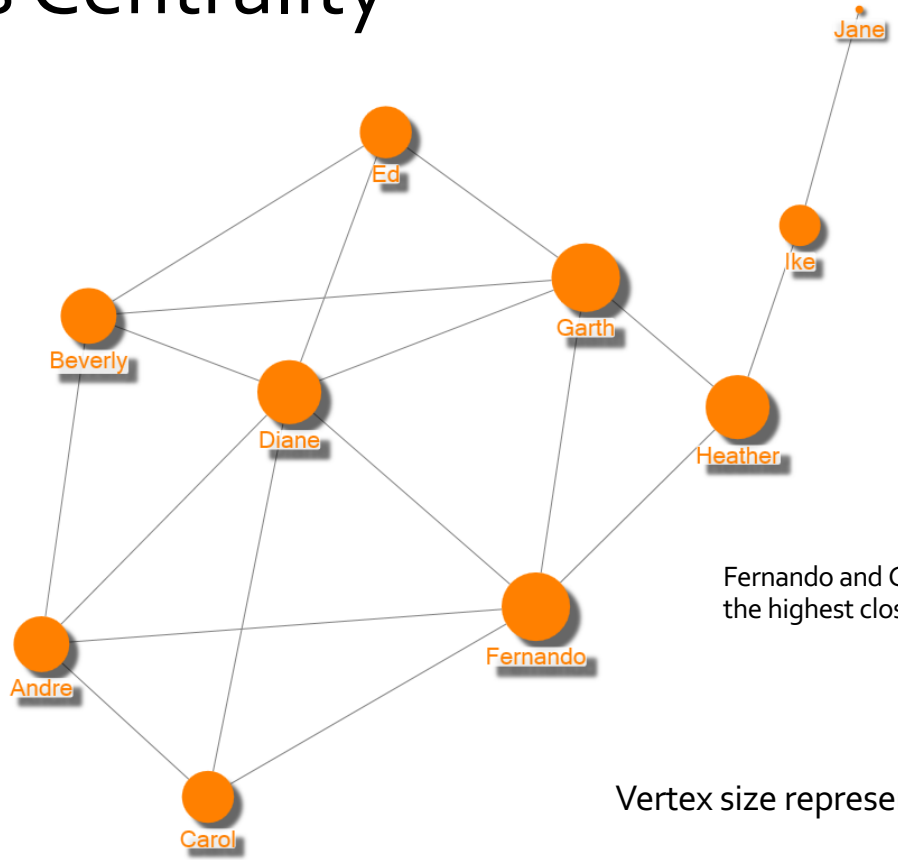
Closeness Centrality

- Measures how close a person is to the others
  - how fast a message can reach all others from a person
- What is the scenario for a person's message to reach all others in the fastest way?
- The **closeness centrality** of a node:

$$C(v) = \frac{1}{\sum_{u \neq v \in V} d(u, v)}$$

Farness of v:
Total distance between *v* and all other nodes

Closeness Centrality

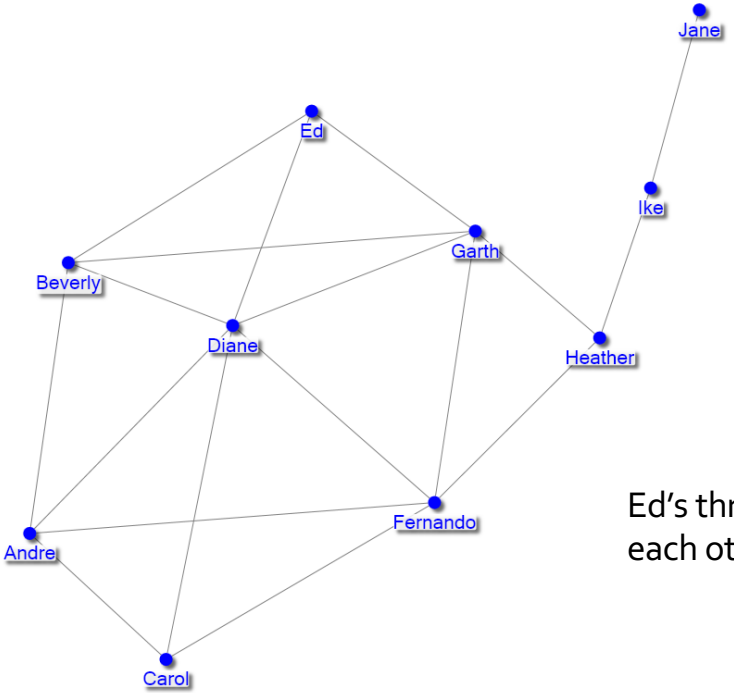


Fernando and Garth are of the highest closeness measures

Vertex size representing closeness

Clustering Coefficient

- Measures how well a person's friends are connected to each other



Ed's three friends know each other

Clustering Coefficient

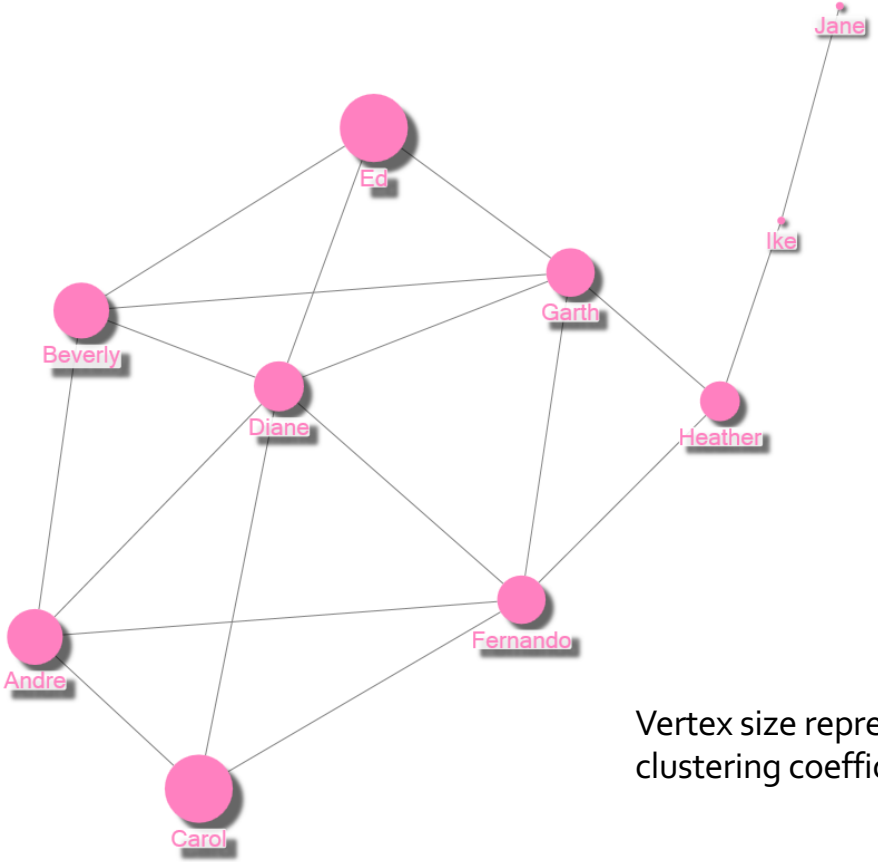
- Clustering coefficient of a node:

$$C(u) = \frac{\text{\# edges connecting } u\text{'s neighbours}}{\text{total \# possible edges connecting } u\text{'s neighbours}}$$

If u has k neighbours, this equals # of edges in a k-vertex clique (complete graph)
$= k\ (k-1)/2$ for an undirected graph
$= k\ (k-1)$ for a directed graph

- Ranged in [0,1]
  - $C(u)$ = 0: no edges among neighbours
  - $C(u)$ = 1: neighbours form a clique

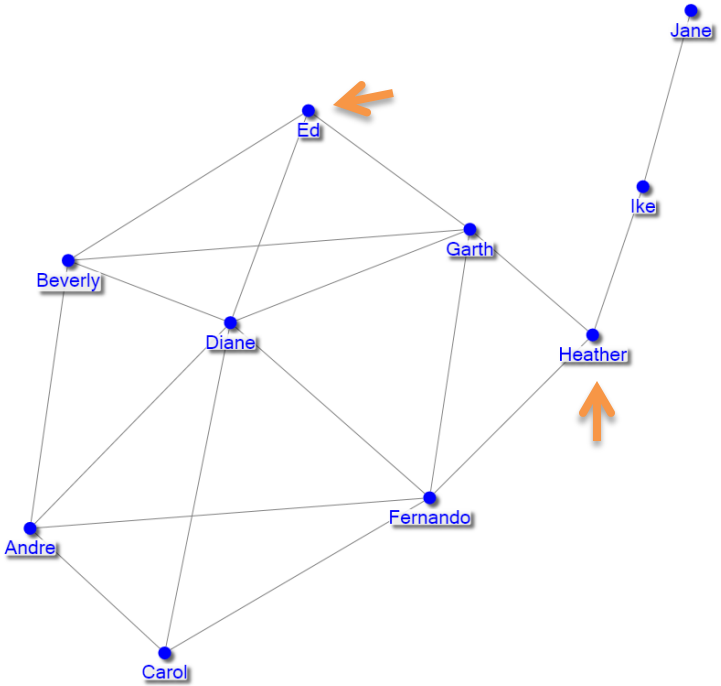
Clustering Coefficient



Vertex size representing clustering coefficient

Eigenvector Centrality

- Measures the influence of a node in a network
  - having a connection to an important person is more important



- deg(Heather) = deg(Ed) = 3
- Ed connects with Diane who is the most popular (i.e., having the largest degree)
- Heather connects to Ike, who is among the least popular
- Hence, Ed's eigenvector centrality is higher

Eigenvector Centrality

- The eigenvector centrality score of a node is:

$$x(u) = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} x(v) = \frac{1}{\lambda} \sum_{v \in V} a_{u,v} x(v)$$

λ : constant

Sum of scores over all u's neighbours

$a_{u,v}$ = 1 iff u, v are neighbours, otherwise $a_{u,v}$ = 0

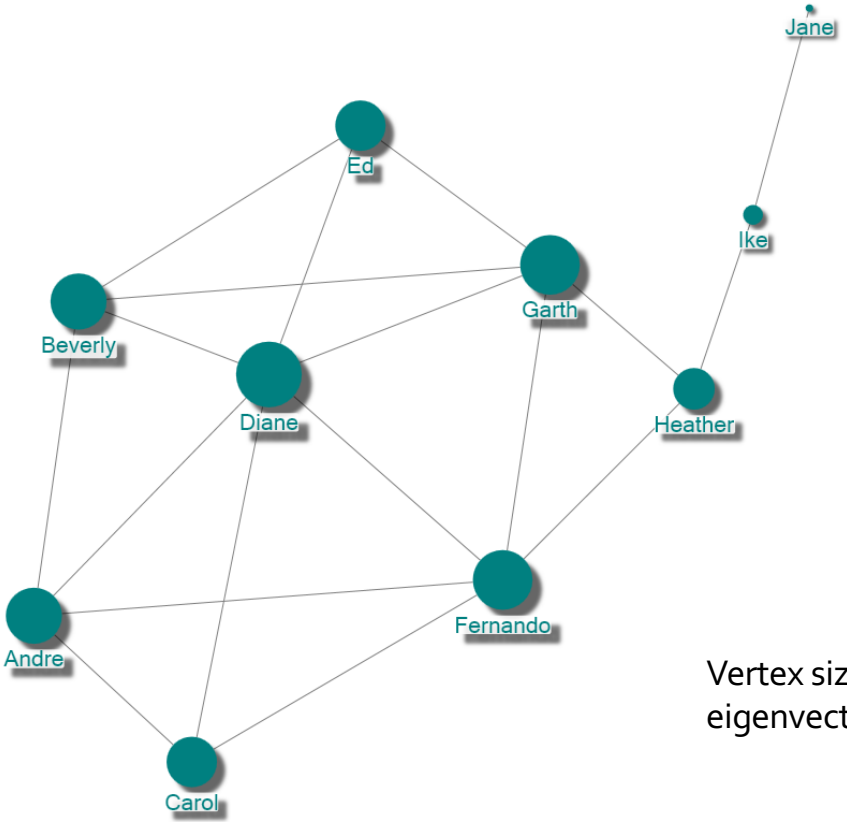
- Written in matrix form gives:

$$\mathbf{A}\mathbf{x} = \lambda \mathbf{x}$$

$\mathbf{A} = (a_{u,v})$: adjacency matrix

Since it is required that all scores (i.e., entries of **x**) be positive,
the eigenvector corresponding to the largest eigenvalue will give us the scores.

Eigenvector Centrality

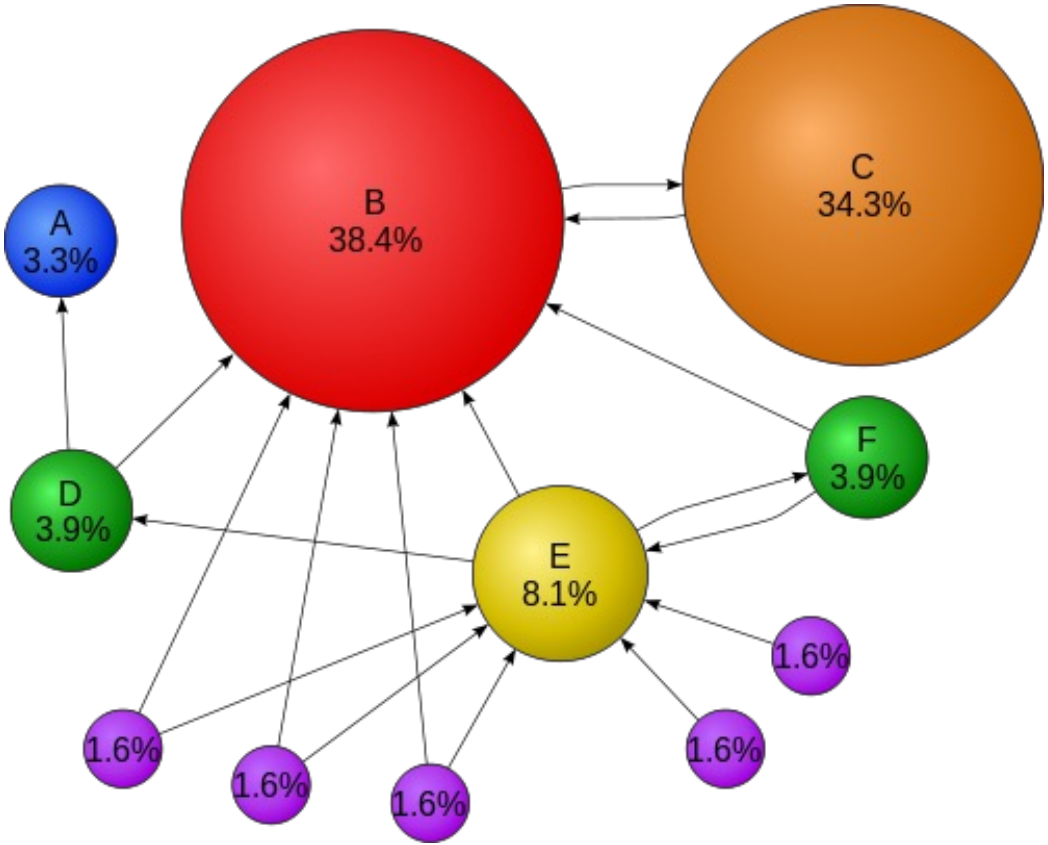


Vertex size representing eigenvector centrality

PageRank

- Used by Google search engine to rank websites in their search results.
- Idea: More important websites are likely to receive more links from other websites.
- A variant of eigenvector centrality
- **Score of a page** = the probability of being brought to a page after many clicks.

PageRank



http://en.wikipedia.org/wiki/PageRank

- The percentage shows the likelihood that a page can be reached after many clicks.
- Assumption: A user start on a random page, and has 85% chance of clicking randomly on any link in the page, and **15% chance of jumping randomly** to any other page in the WWW.

PageRank

- An assumption that there is possibility a surfer will stop following links and jump instead to a random page

$$x(u) = p \sum_{v \in V} a_{v,u} \frac{x(v)}{L(v)} + \frac{1-p}{N}$$

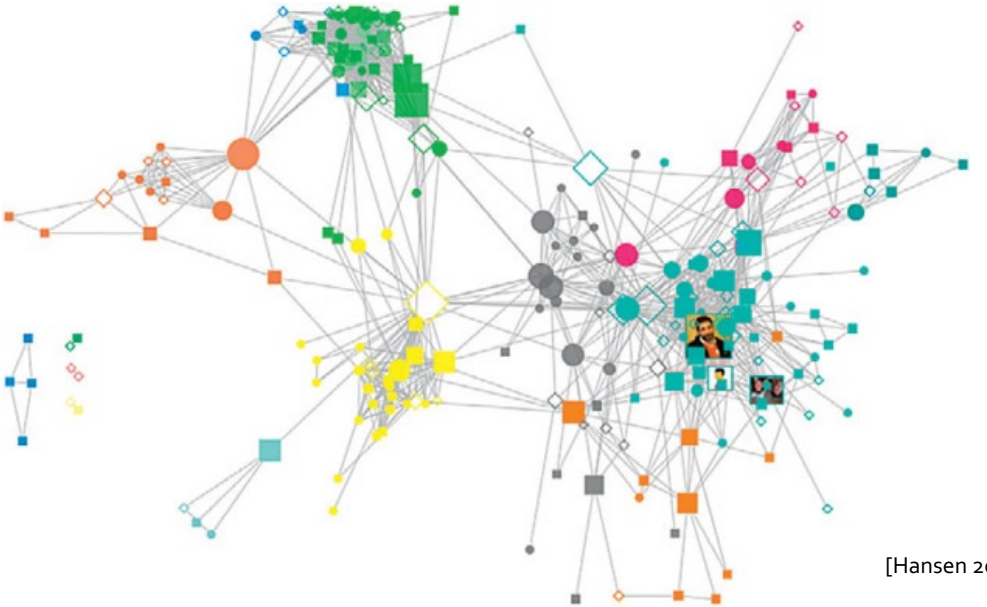
p : probability of following a link
$1-p$: probability of jumping to a random page
N : total number of nodes
$L(v)$: out degree of v

- Can be used for ranking nodes in a general network.

Visual Attributes Mapping

- Map ordered visual attributes to ordered data and unordered visual attributes to categorical data
- Ordered data: Node degree, centrality
  - Ordered visual attributes: Size, line width, opacity
- Categorical data: Gender, affiliation
  - Unordered visual attributes: Color, shape

Visual Attributes Mapping



[Hansen 2011]

shape ↔ gender
color ↔ cluster

size ↔ betweenness
opacity ↔ eigenvector centrality

Tools and Datasets

- Tools for Graph Visualization
  - Gephi
    https://gephi.org
  - NodeXL
    http://nodexl.codeplex.com
- Large Network Datasets
  - A Citation Network Dataset
    http://arnetminer.org/citation
  - Stanford Network Analysis Project
    http://snap.stanford.edu

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